

A deep learning framework for heart rate estimation from facial videos

Gee-Sern Jison Hsu^{a,*}, Rui-Cang Xie^a, ArulMurugan Ambikapathi^b, Kae-Jy Chou^a

^a National Taiwan University of Science and Technology, Taipei 10607, Taiwan

^b Institute for Infocomm Research, Singapore 138632, Singapore

ARTICLE INFO

Article history:

Received 21 July 2019

Revised 21 April 2020

Accepted 3 July 2020

Available online 14 July 2020

Communicated by Zhen Lei

Keywords:

Heart rate estimation

Convolutional Neural Network

Time-Frequency Representation

Facial heart rate database

ABSTRACT

Accurate heart rate is vital to acquiring critical physical data of human subjects. For this reason, facial video-based heart rate estimation has recently received tremendous attention owing to its simplicity and convenience. However, its accuracy, reliability and computational complexity have yet to reach the desired standards. In this work, we have endeavored to develop a novel deep learning framework for real-time estimation of heart rates by using an RGB camera. Our approach consists of the following four steps. We begin Step 1 by detecting the face and facial landmarks in the video to identify the required facial Region of Interests (ROIs). In Step 2, we extract the sequence of the mean of the green-channel video from the facial ROIs, and explore a three-stage sequential filtering, including illumination rectification, trend removal and signal amplification. In Step 3, the Short-Time Fourier Transform (STFT) is employed to convert the 1D filtered signal into the corresponding 2D Time-Frequency Representation (TFR) for characterizing the frequencies over short time intervals. The 2D TFR allows the formulation of the heart rate estimation as a video-based supervised learning problem, which can be solved by exploring a deep Convolutional Neural Network (CNN), as is carried out in Step 4. Our approach is one of the pioneering work that proposes a deep learning framework with TFRs as input for solving the heart rate estimation from facial videos. In addition, we have developed a heart rate database, named the Pulse From Face (PFF), and used it along with the existing PURE (Pulse Rate) database [1] to train the CNN. The PFF database is released for research purpose with this paper. We have evaluated the proposed framework on the MAHNOB-HCI database [2] and the VIPL-HR database [3] and compared its performance with that of other contemporary approaches to demonstrate its efficacy.

© 2020 Published by Elsevier B.V.

1. Introduction

Cardiac pulse rate (aka heart rate) is the most crucial vital sign of a human. Measuring heart rates for diagnosis and prognosis has been standard practice since ancient times. In recent years, remote monitoring of heart rates has become a primary necessity and a core research area. Applications include telemetry monitoring of elderly people, realtime monitoring of premature newborn babies, critical patients, the athletes involved in sports activities and others. In all these cases, it is almost impossible for a trained physician to be physically present at all times and to monitor the heart rates of her patients. To circumvent such situations, heart rate estimation from facial videos has been increasingly investigated. The rationale behind heart rate measurement from facial video sequences is that the blood flow variation across the facial region, although invisible to the human eyes, can in fact be captured by

the videos. As the facial video contains a broader spectrum of reflected light, the problem can be formulated by extract the color variation frequencies correlating them with the heart rates; the analysis of the extracted frequencies can then be used to estimate the heart rates.

An early work was reported by Verkruysse et al. [4], and followed up by several researchers with improvements [5–10]. In [5], Poh et al. applied a blind source separation technique, i.e., independent component analysis (ICA), to the spatially averaged R, G, and B components of the facial image sequence, extracted the components related to cardiac pulses, and calculated the temporal frequency peaks as the heart rate. This method was improved in [6], where machine learning techniques were invoked to enhance the ICA based approach. The issue of interferences caused by illumination changes was addressed by Haan and Jeanne [7], so as to increase the accuracy of the estimated pulse rate. By the measurement and amplification of subtle head motion caused by the Newtonian reaction to the influx of blood at each beat, Balakrishnan et al. [8] tracked the features on the head and extracted the

* Corresponding author.

E-mail address: jison@mail.ntust.edu.tw (G.-S.J. Hsu).

component that best corresponds to heartbeats based on its temporal frequency spectrum.

More recently, the non-rigid motions and illumination interference in the facial videos were jointly considered by Li et al. [9], thereby yielding a better estimate of the heart rate, even under mild physical motions of the subjects. The self-adaptive matrix completion proposed by Tulyakov et al. [10] dynamically selects the face regions most suitable for heart rate estimation and reaches a better accuracy under the condition of mild movements and facial expressions. Cheng et al. [11] introduced an illumination-robust framework using the joint blind source separation (JBSS) and ensemble empirical mode decomposition (EEMD) to evaluate heart rate from videos. The framework is based on the hypotheses that both facial ROI and background ROI have similar illumination variations. The background ROI is considered as a noise reference to denoise the facial signals by using the JBSS. The reconstructed illumination-resisting green channel of the facial ROI is decomposed into intrinsic mode functions using the EEMD to estimate the heart rate. Chen et al. [12] put forward an end-to-end system for video-based measurement of the heart and breathing rates using a CNN. The system uses a motion representation and an attention mechanism to guide the estimation of heart and breathing rates. Niu et al. [3] propose a spatial-temporal representation for the heart rate and design a general-to-specific transfer learning strategy to train a deep heart rate estimator from a large volume of synthetic rhythm signals and a limited number of available face video data. Chen et al. [13] surveyed video-based heart rate measurement approaches, with a focus on recent advancements that deal with illumination variations and motion artifacts. The selected up-to-date methods are summarized with pros and cons under different conditions.

The framework that we propose for facial heart rate estimation consists of the following four steps:

1. Localization of the region of interest (ROI) for extracting the facial color signal;

2. Filtering of the facial color signal so that the part correlated with the heart rate can be extracted;
3. Conversion of the 1D filtered facial color signal to 2D Time-Frequency Representation (TFR);
4. Training a deep convolutional neural network (CNN) to estimate the heart rates given the TFRs as the input sequence.

The configuration of the proposed framework is illustrated in Fig. 1. We first detect the face and the associated facial landmarks, from which we locate the desired facial region of interest (ROI) in each frame. Given that the video captures the color variation caused by arterial blood flow in the face, and it is empirically verified [4,9] that the Green (G) channel contains the strongest plethysmographic signal among the three color channels, we extract the G-channel signal from the facial ROI for the subsequent processing. A three-stage sequential filtering approach is then explored to extract the heart-rate-correlated part from the G-channel signal. The first stage is for illumination rectification, which aims to remove the temporal illumination variation caused by the ambient lights. The second stage is to detrend the signal in order to remove the undesired systematic variation in the signal. The third stage is for signal amplification so that the amplitudes of higher frequency components can be properly enlarged insofar as such components tend to diminish when transformed into the frequency domain.

To better observe the frequency variation in the 1D signal over time, we apply the Short Time Fourier Transform (STFT) to convert the 1D signal to a 2D TFR image. This conversion proceeds in a recursive way, i.e., as the 1D signal is constantly updated for a short time interval, the corresponding 2D TFR is also updated, thus providing a continuous TFR sequence. The STFT is used to determine the local sinusoidal frequency and phase content of a signal as it changes over time. The TFR sequence is then fed into a VGG (Visual Geometry Group) network [14] to estimate the heart rate. To train the VGG network, we extract the TFR image sequences from a training set which is composed of the facial videos from the PURE

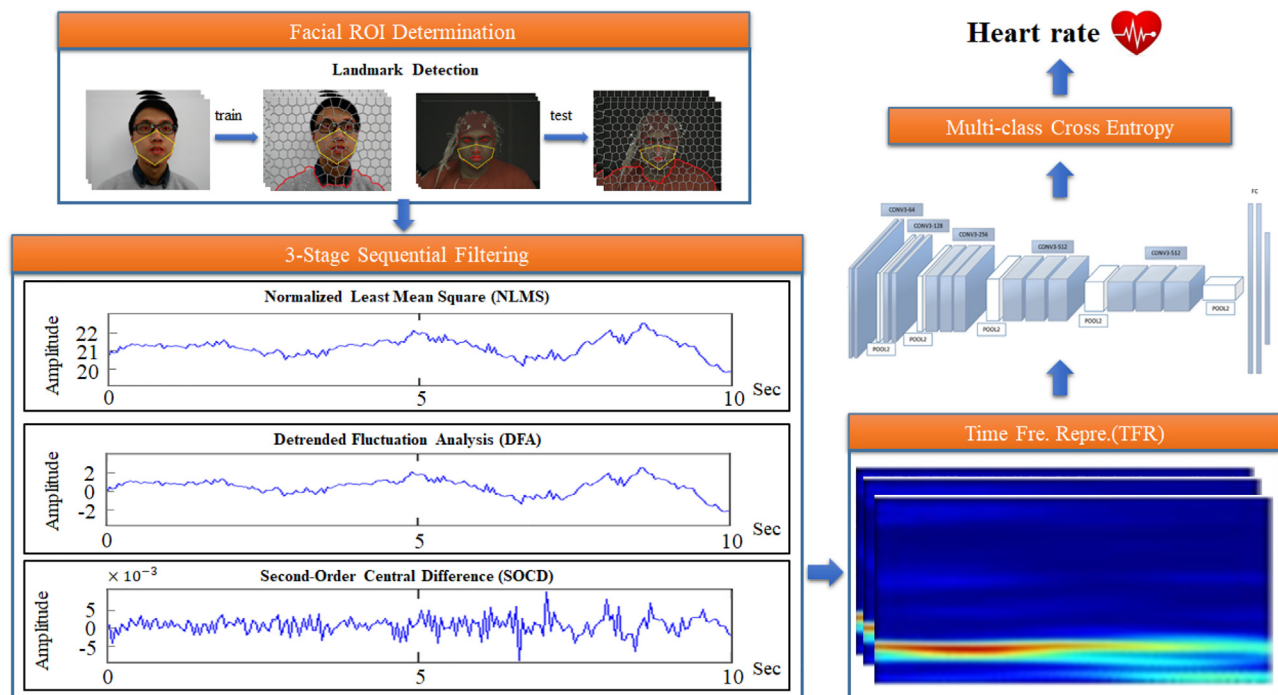


Fig. 1. The work flow of the proposed framework for estimating the heart rate with four modules, namely the facial ROI extraction, 3-stage sequential filtering, time-frequency representation and cross-entropy estimation using VGG-16.

database [1] and the Pulse From Face (PFF) database that we made for this study. The trained VGG network is evaluated on the MAHNOB-HCI database [2] and the VIPL-HR database [3] for a performance comparison with other state-of-the-art methods.

The contributions of this research can be summarized as follows:

1. This is the first framework that fuses the TFR and a CNN, suffices the heart rate bandwidth in the training set, and explores the CNN-based weighted average for the heart rate estimation;
2. The 3-stage sequential filtering is advanced from previous techniques with modifications and a better processing structure;
3. The proposed framework is experimentally verified as performing better than many state-of-the-art approaches;
4. The release of the PFF (Pulse From Face) database, which consists of videos taken under various settings, offers a wide range of characteristics of facial heart rate data that can help advance related research.

This paper is an extension of the work reported in our conference version [15]. The extension includes the following parts:

1. We did not consider the removal of the illumination noise caused by the ambient light in the previous version [15], but it is effectively handled in this extension, as addressed in Section 2.2.
2. The SEEDS (Superpixels Extracted via Energy-Driven Sampling) algorithm is incorporated to segment the input image into the facial skin region, non-skin body region and the background region. This segmentation is crucial for the removal of the ambient-light caused illumination noise.
3. The chrominance-based processing in the previous version is removed in this extension due to inconsistent outcomes of extensively run experiments.
4. In addition to the removal of the chrominance-based filtering, the original filtering in the previous version is completely revised with two major modifications: 1) the previous parallel processing is replaced by the current 3-stage sequential filtering, as presented in Section 3; and 2) the incorporation of illumination rectification into the sequential filtering.
5. More experiments were conducted for this extension to verify the performance of different settings in the 3-stage sequential filtering.

The rest of the paper is organized as follows: The methods for face and facial landmark detection is presented in Section 2, with the superpixel-based approach for the segmentation of an input image into the facial skin region, non-skin body region and background region. Section 3 describes the 3-stage sequential filtering that extracts the heart rate signal from the facial video. The conversion of the 1D heart rate signal into the 2D TFR is given in Section 4, along with the settings of the VGG-16 network for solving the heart rate estimation by visual recognition. The experiments for performance evaluation and a comparison with other contemporary approaches are reported in Section 5, followed by a conclusion with future research directions given in Section 6.

2. Face and facial landmark detection for image segmentation

The first step is the face detection and facial landmark localization, which are crucial for two objectives: 1) Removal of the noises caused by the cross-frame facial ROI jittering; and 2) Segmentation of the facial skin region and non-skin body region. The segmentation of skin and non-skin regions is needed for the subsequent illumination rectification addressed in Section 3. For high performance face detection, we implement the Single Shot multibox Detector

(SSD) [16], which is one of the latest real-time object detectors. As several off-the-shelf facial landmark detectors, e.g., the Regressing Local Binary Feature (RLBF) [17] and the 3D Dense Face Alignment (3DDFA) [18], failed to locate consistent landmarks across frames, we explored the Double Conv-Dropout (DCD) network, which follows the multiple dropout framework that we proposed in Ref. [19] to efficiently prevent overfitting and simultaneously improve the landmark localization accuracy. The cross-frame landmark consistency is ensured by the relatively shallow network configuration. The improved cross-frame landmark consistency suppresses the cross-frame landmark jittering, and in turn reduces the noises in the desired facial color signals.

To the best of our knowledge, all previous studies on heart rate estimation use off-the-shelf facial landmark detectors. For example, Li et al. [9] use the DRMF [20], Tulyakov et al. [10] use the IntraFace [21], and Xuesong Niu et al. [3] explore the SeetaFace [22]. Additionally, none of previous work reports the influences of the cross-frame landmark jittering on the heart rate estimation.

2.1. Single Shot multibox face Detector

We have tested several state-of-the-art landmark detectors, and none of them can generate sufficiently consistent facial landmarks across frames. The cross-frame inconsistent landmarks are caused by the variation of the facial regions given by the face detector across frames. The cross-frame landmark inconsistency induces the cross-frame landmark jittering that makes additional noises to the facial regions from which we extract the signals for heart rate estimation. Both the SDM and RLBF employ the Viola Jones face detector, which leads to cross-frame fluctuation on the detection bounding boxes, making the detected landmarks drift across frames. To circumvent this issue, we propose a customized face and landmark detection framework, as shown in Fig. 2. Our framework fuses the face detection and landmark localization in a unified framework. The cross-frame consistency of the landmark localization is substantially improved when the landmark detector is trained and tested on the faces which are detected and cropped by the associated face detector.

The SSD (Single Shot multibox Detector) [16] begins with a set of default bounding boxes over different aspect ratios and scales per feature map location. In the training phase, a score is generated for the presence of each object category in each default bounding box, and then regression is applied to the default bounding box to better match the bounding box of a ground-truth object. Furthermore, this phase combines predictions from multiple feature maps with different resolutions to handle objects of different sizes. The SSD outperforms many previous real-time detectors because of the following characteristics: 1) a small convolution filter employed to predict the object categories and the offsets in the bounding box locations, 2) separate filters applied for the detection of objects of different aspect ratios, and 3) the multi-feature maps built for multi-scale detection. See [16] for further details.

We customize the SSD for face detection by training it on the WIDER FACE database, which offers 393,703 labeled faces in 32,203 images with a large variability in pose, illumination, expression and scale [23]. We manually removed 13,726 faces which are small (<16 pixels) and poor in quality from the training set. This removal made our customized SSD face detector attain an AP (Average Precision) of 91.8% (0.6% better than the case without this removal), better than the contemporary DPM (90.2%) [24] and the HeadHunter (89.6%) [24] and YOLO-9000 (90.6%) [25].

2.2. Landmark localization and region segmentation

Given an SSD-detected face, the landmark detector has to locate the coordinates of 68 landmarks at specific locations on the

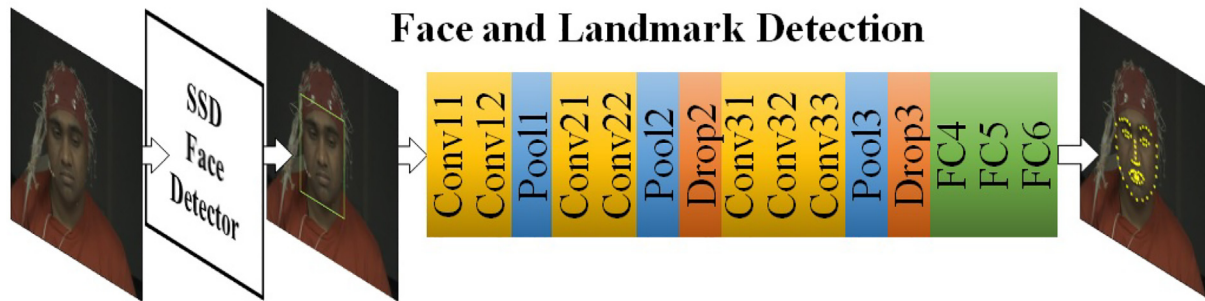


Fig. 2. The proposed face detection and landmark localization network, which employs the SSD face detector and the Double Conv-Dropout (DCD) network for facial landmark detection.

detected face. The landmark detector explores the Double Conv-Dropout (DCD) network that follows the multiple dropout setup that we introduced in [19]. The detector is made of two double-convolution blocks, one triple-convolution block and three fully-connected layers. Each double-convolution block has two convolutional layers followed by a max-pooling layer, and the triple-convolution block has three convolution layers also followed by a max-pooling layer. The double dropouts are implemented at the second double-convolution block and at the triple-convolution block, both next to the max-pooling layers. The loss considered in the landmark detector is the Euclidean distance between the predicted locations and the ground-truth locations. According to the experimental verification in [19], the double dropouts implemented at the convolution layers can better stabilize the network training than the common practice of implementing dropouts at the fully connected layers. We train the landmark detector on the 300 W training set and evaluate the performance on the 300 W testing set [26]. Compared with several contemporary approaches (including the SDM [27], RLBF [17] and 3DDFA [18]), our DCD landmark detector yields 5.12 mean square error (MSE) on the 300 W common set, which is better than that of SDM (5.70), RLBF (5.16) and 3DDFA (5.53).

We use the detected landmarks to segment an input image into the facial skin region, the non-skin body region and the background region. The facial region located by the landmarks are used to initialize the SEEDS (Superpixels Extracted via Energy-Driven Sampling [28]) algorithm for image segmentation. The SEEDS segments the image into skin region and non-skin region, and further segments the non-skin region into body region and background region. We need to obtain the ambient illumination intensity on the non-skin body region so that it can be removed from the illumination on the facial skin region. The core part of the SEEDS is

an energy function minimization using a fast gradient descent algorithm. The energy function minimization amounts to enforcing color similarity across the boundaries and superpixel color histograms. With the initial superpixel partition made by the facial landmarks, the SEEDS continuously refines the superpixels by clustering the pixels with similar color histograms and modifying the superpixel boundaries. Considering the superpixels close to the upper corners of the image can be taken as background superpixels, and those below the chin and close to the bottom of the image can be taken as body superpixels, the refined superpixels can be used to segment the image into the three aforementioned regions. This algorithm reaches a desired compromise between accuracy and efficiency.

The reason for choosing the non-skin body region instead of the background region is that its illumination intensity is a better approximation to the ambient illumination intensity on the face than that of the background region. As the illumination intensity on a surface weakens with the distance to the light source, and the background in an image can have a different distance to the lights, the illumination intensity on the background can be quite different from that on the face. Fig. 3 shows the initial segmentation by the facial ROI and the finalized superpixel-based segmentation. This ROI segmentation can be replaced by other segmentation algorithms as long as the accuracy and speed meet the real-time requirements.

As previous studies have revealed that the green (G) channel of the facial video yields better accuracy in estimating the heart rate than either the red and blue channels [4,9], we only extract the G-channel signals for the subsequent analysis. We obtain the mean of all G-channel pixels inside the facial region in each frame, and accumulate the means over successive frames in a buffer to obtain



Fig. 3. (A) Initialization of the SEEDS segmentation by the facial landmarks; (B) The finalized superpixel-based segmentation.

a 1D time series of the G-channel means. This 1D time series of the G-channel means is abbreviated as *G-mean*, in the following sections.

3. Three-stage sequential filtering

With the *G-mean* continuously registered to the buffer, we perform 3-stage sequential filtering for illumination rectification, detrending and signal amplification. The illumination rectification aims to remove the illumination variation caused by the ambient light. The detrending aims to remove the systematic changes, known as *trends*, in the *G-mean*. The signal amplification amplifies the amplitudes of higher frequency components in the *G-mean* so that the major frequency components can be better detected. The objective of this 3-stage sequential filtering is to extract the heart-rate-correlated (or plethysmographic-dependent) signals for the subsequent processing.

3.1. Illumination rectification

The illumination rectification aims at removing the illumination fluctuation caused by the ambient light. The approach proposed in [9] is referenced but substantially improved and applied in this step. The improvements are made from two aspects. One is the segmentation of the facial skin region from which we extract the skin color variation, and the segmentation of the non-skin body region from which we extract the illumination variation caused by the ambient lights. We assume that the skin color captured by the camera follows the additive rule, where the *G-mean* signal is the sum of the heart-rate-correlated G component and the G component of the ambient light. Let the *G-mean* signal in the buffer be denoted as $[g_f(n)]_{n=1}^{N_b}$, where n is the discrete time index and N_b the buffer size. $g_f(n)$ is composed of the heart-rate-correlated G component or the so-called *illumination rectified G-mean*, denoted as $g_{rec}(n)$, and the G component caused by the ambient light, denoted as $g_a(n)$, i.e.,

$$g_f(n) = g_{rec}(n) + g_a(n) \quad (1)$$

As the ambient light also affects the non-skin region which does not have the heart-rate-correlated G-component, we assume that the G-component collected from the non-skin region is proportional to the G-component of the ambient light, i.e.,

$$g_a(n) = h(n) * g_{ns}(n) \quad (2)$$

where $g_{ns}(n)$ denotes the *G-mean* in the non-skin region, and $h(n)$ is the time-varying proportional coefficient between $g_a(n)$ and $g_{ns}(n)$. (1) can therefore be rewritten as

$$g_f(n) = g_{rec}(n) + h(n) * g_{ns}(n) \quad (3)$$

A similar formulation to (1) and (2) can be found in [9], insofar as the authors employ an iterative Normalized Least Mean Square (NLMS) adaptive filter to solve for $[g_{rec}(n)]$ given $[g_f(n)]$ and $[g_{ns}(n)]$. Our experiments show that the direct application of the NLMS filter requires long sequences of $[g_f(n)]$ and $[g_{ns}(n)]$ stored in the buffer. The required $[g_f(n)]$ and $[g_{ns}(n)]$ sequences must contain 100 data or more available in the buffer (corresponding to 4 s at frame rate 25 FPS). To alleviate this requirement, we propose the following better alternative to reduce the size of the buffered data to 25–50 (1–2 s).

Assuming that $h(n)$ is a slow-varying process as both $g_a(n)$ and $g_{ns}(n)$ are caused by the same ambient light, we can approximate (3) as $g_f(n) = g_{rec}(n) + h_r * g_{ns}(n)$, where h_r is a constant for the sample interval $n_r \leq n \leq n_r + N_r$, and N_r is the interval length.

The constant h_r can be estimated by minimizing $\|g_f(n) - h_r * g_{ns}(n)\|_{n_r \leq n \leq n_r + N_r}$,

$$h_r^* = \arg_{h_r} \min \|g_f(n) - h_r * g_{ns}(n)\| \quad (4)$$

Once the initial $h(n)$ is given as a sequence of h_r over successive sample intervals N_r , the rectified G-mean $g_{rec}(n)$ can be computed by

$$g_{rec}(n) = g_f(n) - h(n) * g_{ns}(n) \quad (5)$$

The iterative Normalized Least Mean Square (NLMS) adaptive filter [9] can then be used to estimate the optimal $h(n)$ as follows:

$$h(n_0 + 1) = h(n_0) + \mu \frac{g_{rec}(n_0)g_{ns}^H(n_0)}{g_{ns}^H(n_0)g_{ns}(n_0)}, \forall n_0 \quad (6)$$

where μ is the step size, H denotes Hermitian operation and $h(0) = 0$. A sample set of the original signal $g_f(n)$, the ambient G component $g_{ns}(n)$ and the rectified G-mean $g_{rec}(n)$ is shown in Fig. 4.

To further improve $g_{rec}(n)$, we then employ a sequential filtering scheme that is composed of detrended fluctuation filtering for bias removal and second order central difference filtering for signal amplification.

3.2. Detrended Fluctuation Analysis

The Detrended Fluctuation Analysis (DFA) is a technique for determining the statistical self-affinity of a signal [29]. It can be applied to remove the systematic fluctuation sometimes observed in a time series. We directly apply Hardstone et al.'s [30] algorithm, which can be summarized as follows:

1. Compute the cumulative sum of the rectified G-mean $g_{rec}(n)$, denoted as $g_{cum}(n)$.

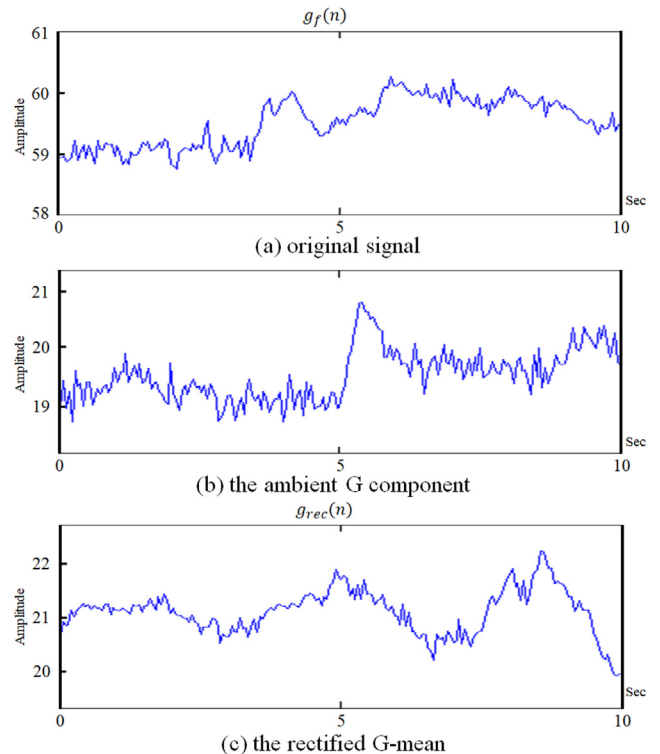


Fig. 4. The revised iterative Normalized Least Mean Square (NLMS) adaptive filtering; the top shows the G-mean $g_f(n)$ collected from the facial region, the middle is the G-mean $g_{ns}(n)$ collected from the non-skin body region, and the bottom is the rectified G-mean $g_{rec}(n)$.

2. Define a set of window sizes, $S_w = [s_j]$, which are equally spaced on a logarithmic scale between the lower bound of an admissible number of data (a common choice is 4 [30]) and the size of the buffer for storing $[g_{rec}(n)]$. For each window size $s_j \in S_w$,
 - Split $g_{cum}(n)$ into a set of windowed data of length s_j with 50% overlap. This set is denoted as $W_g^{(s_j)}$.
 - For each windowed data $w_i^{(s_j)} \in W_g^{(s_j)}$, remove the linear trend (obtained by using a least-squares fit) to create the detrended data $\hat{w}_i^{(s_j)}$.

The above processing can be continued to compute the fluctuation function using the mean of the standard deviations for all identically sized windows. See [30] for details. We use the code available on <https://github.com/dokato/dfa> for the implementation. Fig. 5 shows a segment of the rectified G-mean, $[g_{rec}(n)]$, and after detrending, it is denoted as $[g_{r,d}(n)]$.

3.3. Signal amplification

The derivative of a sinusoid of frequency f is a phase-shifted sinusoid of the same frequency with an amplitude proportional to f . The derivative of a periodic signal that contains several sinusoidal components of different frequencies will still contain those same frequencies, but with altered amplitudes and phases. The derivative makes the components with higher frequencies increased more in amplitude compared to the components with lower frequencies [31]. We apply the second order derivative to the rectified and detrended G-mean $g_{r,d}(n)$ for the amplification of the signal components with higher frequencies, given that the original amplitudes of those components are lower than those of the components with lower frequencies. The second order derivative (SOCD), as shown in the following equation.

$$d(n) = g_{r,d}(n-1) - 2g_{r,d}(n) + g_{r,d}(n+1) \quad (7)$$

We convert $[d(n)]$ from the time domain to the time-frequency domain by applying the short-time Fourier Transform (STFT) with a Hamming Window imposed beforehand. As typical heart rates fall in the range of 41 ~ 240 BPM, we apply a bandpass filter to block the frequencies beyond this range. The conversion of the 1D signal $[d(n)]$ in the time domain to the corresponding 2D representation in the time-frequency domain helps to measure the frequency variation of $[d(n)]$ across a short time interval. Fig. 6 illustrates a sample sequence of the facial G-mean $g_f(n)$, processed

by the above three-stage sequential filtering, resulting in the derivative-amplified data sequence $[d(n)]$. Note that in the following we still refer to $[d(n)]$ as the (processed) G-mean for simplicity.

3.4. Discussion

Although the above three filtering approaches are based on exiting techniques, we would like to highlight the differences from previous work and the novelties as follows:

- Different from our previous work [15], in which we propose a parallel-processing configuration with a chrominance-based filtering component and without illumination rectification, we change it to a sequential-processing configuration. In the ablation study reported in Section 5.3, we experimentally verifies that the new sequential processing is better than the previous parallel processing. Furthermore, the ablation study also shows the contributions of different combinations of the filtering components.
- As presented in Section 3.1, our illumination rectification is a modified version of the techniques employed in [9]. The modification reduces the size of the buffered data needed to generate the illumination-rectified signals.
- Detrending has been implemented in previous research, e.g., [7,9], by using a smoothness priors approach, but we use the DFA algorithm proposed by Hardstone et al. [30] as the latter provides a unique insight into the functional organization of neuronal systems [30]. Our work is the first application of Hardstone et al.'s algorithm to detrend the heart rate signals.

4. Deep learning for heart rate estimation

Given the 3-stage filtered G-mean time series signal $[d(n)]$ in the buffer, we convert it into a 2D Time-Frequency Representation (TFR), formulate the problem as a supervised learning problem, and solve it by using a deep CNN. Our approach consists of a learning phase and a testing phase. In the learning phase, the CNN is trained to learn the heart rate patterns on the TFRs in a training set; in the testing phase, the trained CNN identifies the heart rates on the TFRs from an independent testing set.

The TFR, shown in Fig. 1, is an image that reveals a set of frequencies collected at short time intervals across a specific time span. Specifically, for each buffer length (256 frames), the obtained TFR is a 200×256 image, where 200 corresponds to the heart rate range (41–240 bpm) and 256 corresponds to the time steps for the G-means to register, which covers a 10-s time span as the frame rate is kept at 25 Hz. The consecutive 1D G-means are constantly transformed by the STFT to obtain the 2D TFR. The TFR is designed to be updated every 10 time steps (about 0.4 s) in order to compromise the update speed and the required processing during the update interval. The required processing includes the heart rate estimation by passing the updated TFR through the VGG-16 network for regression.

The parameters and settings of the VGG-16 network [14] exploited are given in Table 1. The network has two double-convolution blocks, three triple-convolution blocks and three fully connected layers. The rectified linear unit (ReLU) is used for activation in the convolution layers, and a max-pooling layer is connected to each convolution block so as to downsample the features. A softmax function is connected to the last fully-connected layer FC8 for computing the cross-entropy loss.

The novelties of the proposed combination of the TFRs and the VGG network can be summarized as follows:

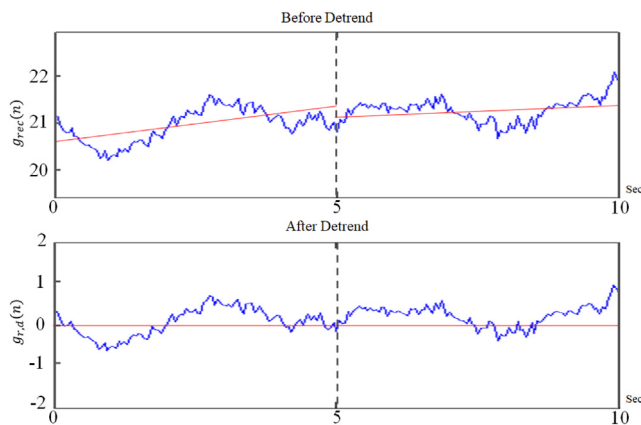


Fig. 5. A segment of the rectified G-mean $[g_{rec}(n)]$ is shown on the top, and the bottom shows the same segment after detrending, denoted as $[g_{r,d}(n)]$.

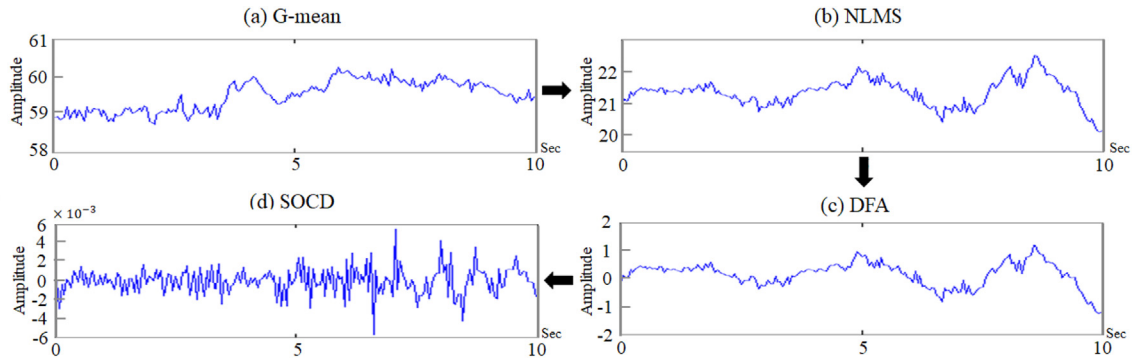


Fig. 6. A sample for demonstrating the input and output throughout the 3-stage sequential filtering. The G-mean shows the G-channel mean $g_f(n)$ as the input, the NLMS shows the illumination rectified $g_{rec}(n)$, the DFA shows the detrended $g_{r,d}(n)$ and the SOCD shows the derivative-amplified $d(n)$ as the output.

Table 1

The parameters and settings of the exploited VGG-16 network.

Layer Type Name	0 Input –	1 Conv C1_1	2 ReLU R1_1	3 Conv C1_2	4 ReLU R1_2	5 MPool Pool1	6 Conv C2_1	7 ReLU R2_1	8 Conv C2_2	9 ReLU R2_2	10 MPool Pool2	11 Conv C3_1	12 ReLU R3_1
Support	–	3	1	3	1	2	3	1	3	1	2	3	1
Filt dim	–	3	–	64	–	–	64	–	128	–	–	128	–
Num filts	–	64	–	64	–	–	128	–	128	–	–	256	–
Stride	–	1	1	1	1	2	1	1	1	1	2	1	1
Pad	–	1	0	1	0	0	1	0	1	0	0	1	0
Layer Type Name	13 Conv C3_2	14 ReLU R3_2	15 Conv C3_3	16 ReLU R3_3	17 Max Pool Pool3	18 Conv C4_1	19 ReLU R4_1	20 Conv C4_2	21 ReLU R4_2	22 Conv C4_3	23 ReLU R4_3	24 Max Pool Pool4	25 Conv C5_1
Support	3	1	3	1	2	3	2	3	1	3	1	2	3
Filt dim	256	–	256	–	–	256	–	512	–	512	–	–	512
Num filts	256	–	256	–	–	512	–	512	–	512	–	–	512
Stride	1	1	1	1	2	1	2	1	1	1	1	2	1
Pad	1	0	1	0	0	1	0	1	0	1	0	0	1
Layer Type Name	26 ReLU R5_1	27 Conv C5_2	28 ReLU R5_2	29 Conv C5_3	30 ReLU R5_3	31 Max Pool Pool5	32 FC FC6	33 ReLU ReLU6	34 FC FC7	35 ReLU ReLU7	36 FC FC8	37 Softmax Prob.	
Support	1	3	1	3	1	2	7	1	1	1	1	1	
Filt dim	–	512	–	512	–	–	512	–	4096	–	4096	–	
Num filts	–	512	–	512	–	–	4096	–	4096	–	200	–	
Stride	1	1	1	1	1	2	1	1	1	1	1	1	
Pad	0	1	0	1	0	0	0	0	0	0	0	0	

- Different from most previous work that takes the highest peak in the Fourier domain of the extracted signal to be the heart rate, we consider a sequence of the highest peaks over a short time interval revealed in the TFRs (a sample TFR is shown in Fig. 8) so that our approach can have better robustness against the noises caused by face movements and other variables.
- Although a few latest works exploit the CNNs for heart rate estimation [3,32], our approach is a pioneering work that fuses the TFRs and a CNN to formulate the heart rate estimation as a visual recognition problem. This formulation enables the end-to-end training, the expansion of the heart rate bandwidth in the training set, and the weighted average of the candidate heart rates for improving the overall accuracy.

5. Experiments and PFF Database

In this section, the Pulse From Face (PFF) Database that we made for this study is presented, followed by the performance evaluation with a comparison with other contemporary methods. As our approach is supervised learning, there must be a training set and a testing set. The PFF and the PURE database [1] are used together as the training set, and the MAHNOB-HCI database [2],

which is used in a few contemporary studies for performance assessment [9,7,10], is considered as the testing set. The combination of the PFF and PURE ensures that the training set contains a sufficient bandwidth of heart rates for learning. The PFF has been made available at <https://github.com/HaoRecog/Pulse-From-Face-Database>. In addition, we also evaluate our approach on the latest VIPL-HR database [3]. We follow the fivefold subject-exclusive testing protocol to divide the training and testing sets, and use the average heart rate as the ground-truth heart rate for each video.

5.1. Pulse from face (PFF) database

To make the PFF database, we used a Nikon D5300 camera with frame rate 50 Hz and resolution 1280×720 pixels for recording facial videos. Two Mio Alpha II wrist heart rate monitors, rectified by medical clinicians, were used to record the subjects' heart rates, and the average BPM of the two readings was taken as the ground truth. Each video clip was recorded for 3 min. The subjects were sitting in front of the camera at an average distance of 0.5 meters. Videos were taken under various illumination conditions that were made by randomly turning on and off the indoor fluorescent lights

and partial coverage of ambient lights. To our best knowledge, this type of dynamic illumination variation data is scant in the existing public databases. The PFF also includes different body movements. The videos were collected from 13 healthy subjects whose physical conditions were confirmed normal by medical clinicians. Each subject was asked to face the camera and the videos were collected for the following five scenarios:

1. Bright-Static (BS): All fluorescent lights were on. Each subject sit still without body movements.
2. Bright-Moving (BM): Under the same illumination condition as of the above 1, each subject was directed to move his/her body to the right and left parallel to the camera repeatedly, with frequency kept between 0.2–0.5 Hz.
3. Dark-Static (DS): All fluorescent lights were turned off with ambient light primarily from the windows and computer monitors. Each subject remained still.

4. Dark-Moving (DM): Under the same illumination condition in 3, each subject moved in the same way as described in 2.
5. Bright-Riding (BR): Under the same illumination condition in 1, each subject was riding on an exercise bike at a constant speed.

5.2. Experimental setup

The proposed framework was implemented on a Windows PC with an Intel Core i7-3.6 GHz CPU, 16 GB RAM and NVIDIA GeForce GTX TITAN-X GPU. The VGG-16 network, built on the Caffe [33] framework, was pretrained on ImageNet for generic object recognition [14]. Multithread processing was adopted to handle the heart estimation in one thread and the video processing in the other. We conducted two sets of experiments for the performance evaluation. In the first experiment, we used the PFF and the PURE databases for training, and the MAHNOB-HCI database for testing. In the second experiment, we used the VIPL-HR database [3] and

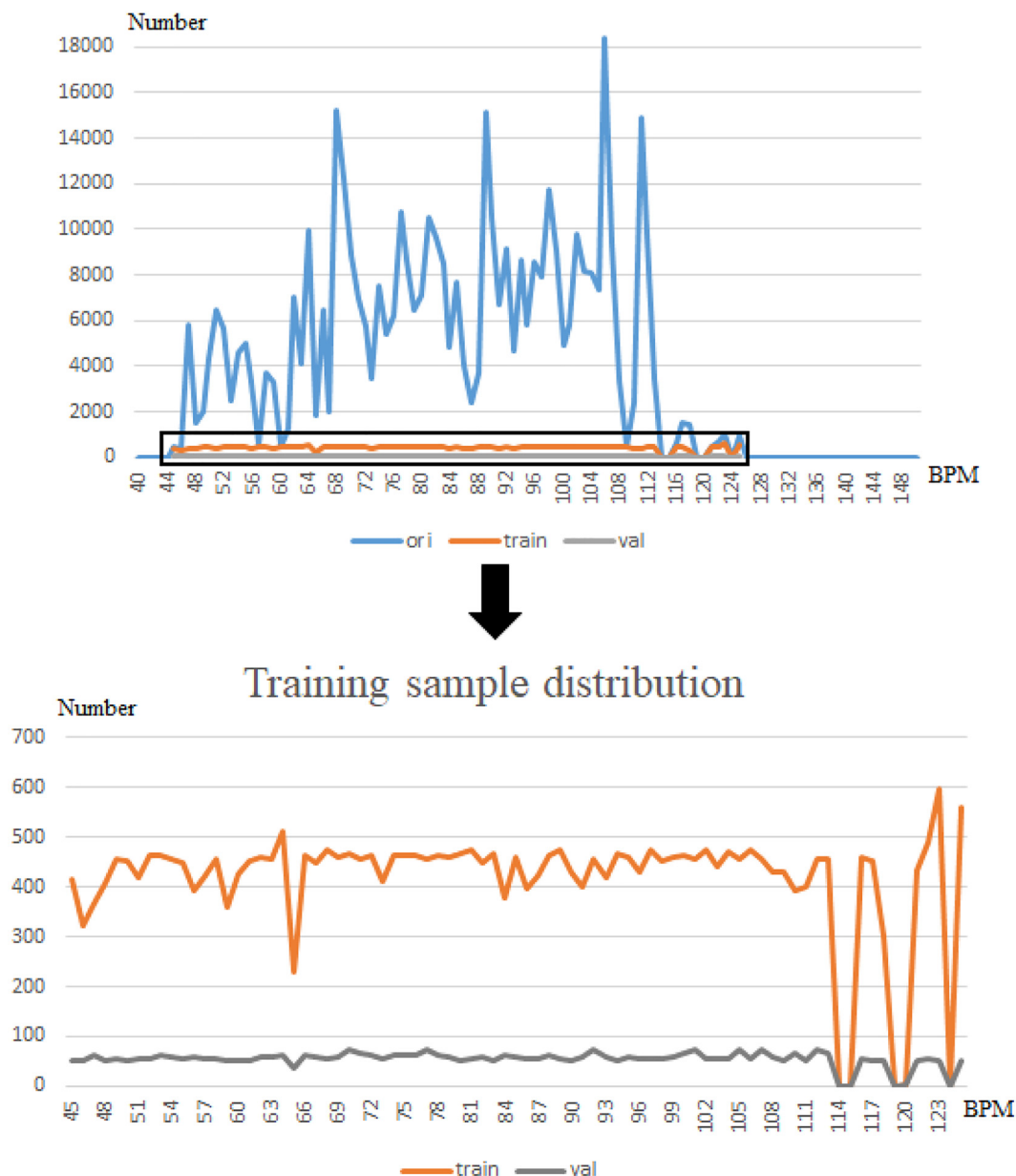


Fig. 7. The number of TFR images at different heart rates (BPM) in the originally collected samples (blue) illustrates severe data imbalance. The partially selected training set (orange) and validation set (gray) were made to be evenly distributed across different heart rates. The chart below is a close-up of the black bound box in the top chart.

followed the training and testing protocol released by the database maker.

5.2.1. Experiments on MAHNOB-HCI

The PURE database consists of 60 video clips collected from 10 subjects and makes available videos with heart rates between 42 and 141 BPM. There are many *singularity* rates in between where no subjects have those heart rates. The PFF consists of 59 video clips collected from 13 subjects with heart rates between 62 and 121 BPM, with many singularity rates in between. The combination of both datasets reduces the singularity rates and offers a better training set. The MAHNOB-HCI database offers 527 videos collected from 27 subjects with facial expression variations and some body movements.

To determine the best hyper-parameters, including the learning rate and momentum, for training the VGG-16, we conducted a subject-independent 10-fold cross-validation test on the training set with 23 subjects (13 from the PFF and 10 from PURE). In each fold, 21 subjects were used for training and the other 2 subjects for testing. In line with the processes described in Sections 3 and 4, each video was processed at 25 FPS and transformed into a sequence of TFR images, and each TFR image was labeled with the heart rate measured by the average of the two wrist heart rate monitors, thus enabling the end-to-end training of the VGG network.

Fig. 7 shows the numbers of TFR images across the working range of heart rates for a case selected out of the 10-fold cross validation. The blue shows the total number of the original TFR images obtained. It can be seen that the number distribution is severely imbalanced. The numbers of the images at certain heart rates, e.g., 70, 90 and 106 BMP, are tens/hundreds of multitudes of those at *rare* heart rates, e.g., 46 and 65 BMP. There are still a few singularity rates, e.g., 114, 115, 119, 120 and 124 BPM. To alleviate the issues with imbalanced data, we selected a subset with the number of images around 450 at each available heart rate for training (shown in orange in Fig. 7), and an independent subset with 70 images at the available heart rate for validation (in gray). We selected the best trained VGG network out of the 10 folds of the cross-validation tests, and retrained it using the overall combined datasets, and then evaluated its performance on the MAHNOB-HCI database.

To better estimate the heart rate and handle the singularity rate issues, we exploited the softmax function at the output layer of the network. The softmax function η takes as input a 200-dimensional feature vector $\vec{x} \in R^{200}$ from the fully-connected layer (FC8), and outputs a 200-dimensional vector $\vec{y} = \{y_i \in [0, 1]\}_{i=1, \dots, 200}$, where y_i is the probability that $\vec{x}$ is classified to Class- i (or the i -th heart rate R_i). It can be written as a normalized exponential as shown below:

$$y_i = \frac{\exp(\vec{w}_i \cdot \vec{x})}{\sum_i \exp(\vec{w}_i \cdot \vec{x})} \quad (8)$$

where $\vec{w}_i \in R^{200}$ is the weight vector of the i -th output neuron. The estimated heart rate R_{es} is considered as the regression of all output probabilities multiplied with the corresponding heart rate (class), which is written as follows:

$$R_{es} = \sum_{i=1}^{200} y_i R_i \quad (9)$$

A case from our experiments with an estimated heart rate of 78 BPM and ground truth 76 BPM is illustrated in Fig. 8, along with the 10 largest probabilities of the heart rates from the softmax output function. It can be seen that the estimated heart rate depends on the first few largest probabilities, while the influence from the remaining low probabilities can be ignored.

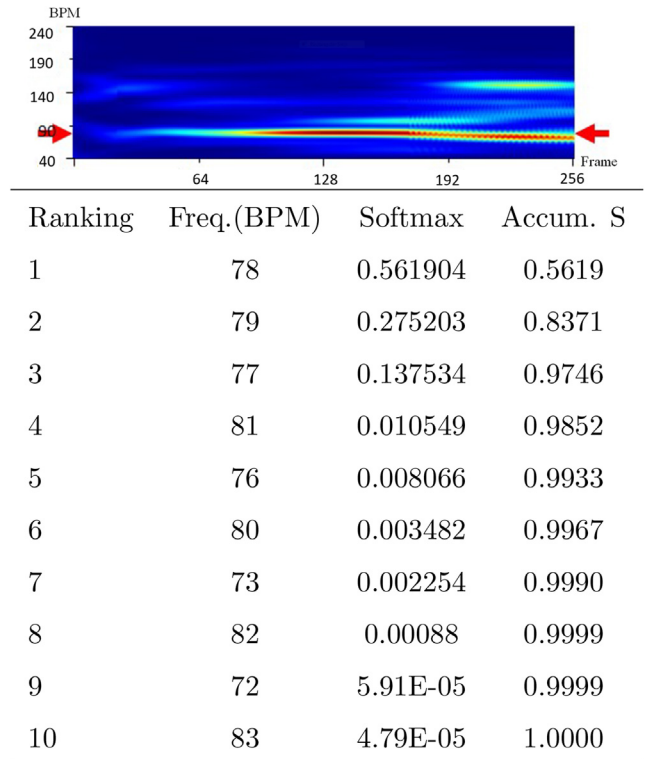


Fig. 8. Top: A TFR sample image; Bottom: The heart rates of the top 10 probabilities from the softmax function of the VGG net. The ground truth is 76 BPM and the estimate according to (9) is 78 BPM.

5.2.2. Experiments on VIPL-HR

The VIPL-HR database [3] contains 2,378 visible light videos and 752 near-infrared videos of 107 subjects. Nine different conditions, including various head movements and illumination conditions, were taken into consideration. All videos were recorded using Logitech C310, RealSense F200 and the front camera of a HUAWEI P9 smartphone, and the ground-truth HR was recorded using a CONTEC CMS60C BVP sensor. The protocol adopted by the database maker was the fivefold subject-exclusive cross validation and the average heart rate was taken as the ground-truth heart rate for each video. We followed the same protocol, and used the model trained on PURE and PFF (same as in the first experiment) as the pre-trained network and trained it on the training set in each fold. Only the visible light videos were considered in our experiments.

5.3. Performance evaluation

The performance for heart rate estimation is commonly reported in terms of the following metrics: the mean error m_e , the error standard deviation σ_e , the root mean square error r_e , the mean of error rate percentage m_{er} and the Pearson correlation ρ . For a better estimation, the first four metrics, m_e , σ_e , r_e and m_{er} , must be lower, and the last ρ must be higher. These metrics are computed based on the estimated heart rate $[h_e(i)]_i$ and the ground truth heart rate $[h_g(i)]_i$ of the samples in the databases, and the definition and associated computations are given in Table 2.

To justify the 3-stage sequential filtering, an experiment was conducted to compare the performance with the illumination rectification only (similar to the work reported in [9]), with the detrending added on but without SOCD, with the SOCD added on but without detrending, and with all three filters processed in sequence. All procedures were the same as described above with the only difference being the time series which was transformed

Table 2
Metrics for assessing performance.

Metric	Definition
$e_h(i)$	$e_h(i) = h_e(i) - h_g(i)$, the error between the estimate $h_e(i)$ and the ground truth $h_g(i)$ for the i -th TFR frame at frame rate 2.5 Hz.
m_e	The mean of the e_h
σ_e	The standard deviation of e_h
e_{rm}	The root mean square error of e_h
m_{er}	The mean of error-rate percentage: $m_{er} = \frac{1}{N} \sum_{i=1}^N \frac{ e_h(i) }{h_g(i)}$
ρ	Pearson's correlation between h_e and h_g : $\rho = \frac{\sum_{i=1}^N (h_e(i) - m_e)(h_g(i) - m_g)}{\sigma_e \sigma_g} m_e$ (m_g and σ_g are the mean and std. of h_g (h_g), respectively).

Table 3
Performance for Illumination Rectification (IR) Only, IR and Detrending (Det), IR and SOCD, and All 3-stage filtering with 3DDFA landmark detector.

Filtering	$m_e(\sigma_e)$	e_{rm}	m_{er}	ρ
Illum. Rec. (IR)	-4.25 (4.65)	3.85	7.0%	0.78
IR+Det	-3.67 (4.79)	3.61	6.7%	0.79
IR+SOCD	-3.26 (5.39)	3.33	6.1%	0.78
IR+Det+SOCD	-2.07 (4.23)	3.08	4.6%	0.84
IR+Det+SOCD ^{3DDFA}	-4.11 (5.26)	3.57	5.8%	0.61

The boldface denotes the best performance in each metric.

into TFR image sequences. The results are shown in Table 3, with the best in each metric being shown in boldface. These results confirm that accuracy can be improved when the three filters are all performed in sequence.

To demonstrate the influences of cross-frame landmark consistency, we ran the same experiment with the 3-stage sequential filtering but using the SSD face detector followed by the 3DDFA landmark detector [18]. The performance, given in the last row in Table 3, clearly shows the deterioration caused by the cross-frame landmark jittering.

Table 4
Performance Comparison with Contemporary Approaches on MAHNOB-HCI (the best three in each metric are shown in boldface).

Method	$m_e(\sigma_e)$	e_{rm}	m_{er}	ρ
Poh et al. [5]	-8.95 (24.3)	25.9	25.0%	0.08
Poh et al. [34]	-2.04 (13.5)	13.6	13.2%	0.36
Bala et al. [8]	-14.4 (15.2)	21.0	20.7%	0.11
Li et al. [9]	-3.30 (6.88)	7.62	6.87%	0.81
Haan et al. [7]	4.62 (6.50)	6.52	6.39%	0.82
Tul. et al. [10]	3.19 (5.81)	6.23	5.93%	0.83
Niu et al. [3]	0.30 (4.48)	4.49	4.37%	N.A.
Hsu et al. [15]	-4.39 (4.65)	4.27	4.95%	0.79
Proposed	-2.07 (4.23)	3.08	4.61%	0.84

Table 5
Confusion matrix of the number of estimated heart rates (Est.) versus that of the Ground Truth (G.T.).

Est.	G.T.				
	41–60	61–70	71–80	81–90	91–125
41–60	15	2	0	0	0
61–70	13	86	7	0	1
71–80	4	33	96	9	6
81–90	1	23	54	87	29
91–125	0	3	5	12	41
Sum	33	147	162	108	77

Table 6
Performance Comparison with Contemporary Approaches on VIPL-HR (the best three in each metric are shown in boldface).

Method	$m_e(\sigma_e)$	e_{rm}	m_{er}	ρ
Tul. et al. [10]	10.8 (18.0)	21.0	26.7%	0.11
Wang et al. [35]	7.87 (15.3)	17.2	18.5%	0.30
Haan et al. [7]	7.63 (15.1)	16.9	17.8%	0.28
Chen et al. [36]	-2.60 (13.6)	13.8	13.6%	0.11
Niu et al. [3]	1.02 (8.88)	8.94	7.38%	0.73
Niu et al. [32]	0.73 (8.11)	8.14	6.71%	0.76
Proposed	-0.74 (9.10)	7.82	9.53%	0.58

Table 4 shows the proposed approach compared with other state-of-the-art methods. Our approach outperforms the other methods in the error standard deviation σ_e , root mean square error e_{rm} and Pearson's correlation ρ , and it is the second best in the error-rate percentage mean m_{er} . Niu et al. [3] claim the best error mean m_e , followed by Poh et al. [34] and then the proposed approach. Although our approach and Poh et al.'s are almost equally good in the error mean m_e (-2.07 and -2.04, respectively), the proposed approach clearly outperforms Poh et al. when considering the error standard deviation σ_e , which are 4.23 and 13.5, respectively.

Note that if the 3-stage sequential filtering is replaced by the illumination rectification only, our filtering part can be considered similar to that of Li et al. [9]. The comparison of Li et al. [9] in Table 4 and the Illum. Rec. in the first row in Table 3 shows that the proposed CNN-based learning outperforms the adaptive filtering in Ref. [9] in σ_e , e_{rm} and ρ , and both are almost equally effective in m_{er} , although it has slightly poorer results for m_e .

To better understand the influences of the training data on the performance, Table 5 shows the confusion matrix revealing how much the estimated heart rates (Est.) deviate from the ground-truth heart rates (G.T.) in discrete ranges. Due to the relative insufficiency of data on both the lower boundary rates (< 46BPM) and higher boundary rates (> 115BPM), the estimates at these boundary heart rates are partially biased and falsely regressed to the neighboring ranges. The falsely regressed parts could be alleviated if more data on the boundary rates is available.

Table 6 shows the proposed approach compared with other state-of-the-art methods on the VIPL-HR database [3]. Because the VIPL-HR database was released after most of the methods in the table were published, the performance of those methods are directly duplicated from [32], which could be obtained by the authors' reimplementation and experiments. The proposed approach performs similarly, as shown in Table 4. In terms of $|m_e|$ and e_{rm} , the proposed approach outperforms almost all others, although the method by Niu et al. [32] appears marginally better for $|m_e|$. As for the other metrics, the proposed approach ranks in the top three.

5.4. Discussion

It has been experimentally verified that the proposed approach outperforms many state-of-the-art methods, and this performance is primarily due to the following modules and techniques explored in our framework:

- Enhanced cross-frame landmark consistency by the customized face and landmark detector: The customized face and landmark detector, as presented in Section 2, effectively removes the cross-frame landmark jittering, and in turn diminishes the noises caused by the unstable facial ROIs across frames.

- Three-stage sequential filtering: The significance and novelties of the three-stage sequential filtering are elaborated in Section 3.4. The three filters are better designed and structured for illumination rectification, detrending and amplification so that the output signals can better capture the heart rates.
- Deep learning on time–frequency representation (TFR) with sufficient-bandwidth training data: As the novelties highlighted in Section 4, this is the first work that fuses the TFR and a CNN, suffices the heart rate bandwidth in the training set, and explores the CNN-based weighted average for the heart rate estimation.

6. Conclusion

We have proposed and experimentally validated a deep learning framework for heart rate estimation. The framework consists of the localization of the facial region of interest, the extraction of the heart-rate-dependent color signals, the conversion of the 1D color signals to 2D TFR image sequences, and the formulation of the heart rate estimation into a supervised learning problem which is solvable by exploiting the VGG network. To have a sufficient bandwidth of the heart rates in the training set, we created the PFF database and now offer it to the research community with this paper. The experiments on the MAHNOB-HCI and VIPL-HR benchmark databases have further verified the efficacy of the proposed framework.

Most skin-based heart rate estimation methods only consider frontal faces, which is a severe limitation for practical applications. We implemented a camera network for taking facial videos from various viewpoints, under different lighting conditions, and with different facial movements and expressions. The data collected by this camera network will be further explored in subsequent research for estimating heart rates from non-frontal and moving face images.

CRedit authorship contribution statement

Gee-Sern Jison Hsu: Conceptualization, Methodology, Investigation, Validation. **Rui-Cang Xie:** Data curation, Software, Validation, Visualization, Investigation. **ArulMurugan Ambikapathi:** Writing - original draft. **Kae-Jy Chou:** Writing - review & editing, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research is partially funded by the Ministry of Science and Technology (MoST), Taiwan, with Grant No. 106-2221-E-011-144-MY3, and by the Center for Cyber-Physical System Innovation under the Higher Education Sprout Project, the Ministry of Education (MOE), Taiwan.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.neucom.2020.07.012>.

References

- [1] R. Stricker, S. Muller, H.-M. Gross, Non-contact video-based pulse rate measurement on a mobile service robot, in: Proc. IEEE International Symposium on Robot and Interactive Communication, 2014, pp. 1056–1062.
- [2] S. Mohammad, J. Lichtenauer, T. Oun, M. Pantic, A multimodal database for affect recognition and implicit tagging, *IEEE Transaction on Affective Computing* 3 (1) (2012) 42–55.
- [3] X. Niu, H. Han, S. Shan, X. Chen, Synrhythm: Learning a deep heart rate estimator from general to specific, in: ICPR, 2018, pp. 3580–3585.
- [4] W. Verkrusse, L.O. Svaasand, J.S. Nelson, Remote plethysmographic imaging using ambient light, *Optics Express* 16 (26) (2008) 21434–21445.
- [5] M.Z. Poh, D.J. McDuff, R.W. Picard, Non-contact, automated cardiac pulse measurements using video imaging and blind source separation, *Optics Express* 18 (10) (2010) 10762–10774.
- [6] M. Hamed, R.A. Calvo, H. Yan, A machine learning approach to improve contactless heart rate monitoring using a webcam, *IEEE Journal of Biomedical and Health Informatics* 18 (4) (2014) 1153–1160.
- [7] G.D. Haan, V. Jeanne, Robust pulse rate from chrominance-based rppg, *IEEE Transaction on Biomedical Engineering* 60 (10) (2013) 2878–2886.
- [8] G. Balakrishnan, F. Durand, J. Guttag, Detecting pulse from head motions in video, in: Proc. IEEE Computer Vision and Pattern recognition (CVPR), 2013, pp. 3430–3437.
- [9] X. Li, J. Chen, G. Zhao, M. Pietikainen, Remote heart rate measurement from face videos under realistic situations, in: Proc. IEEE Computer Vision and Pattern recognition (CVPR), 2014, pp. 4264–4271.
- [10] S. Tulyakov, X. Alameda-Pineda, Self-adaptive matrix completion for heart rate estimation from face videos under realistic conditions, in: Proc. IEEE Computer Vision and Pattern recognition (CVPR), 2016, pp. 2396–2404.
- [11] J. Cheng, X. Chen, L. Xu, Z.J. Wang, Illumination variation-resistant video-based heart rate measurement using joint blind source separation and ensemble empirical mode decomposition, in: *IEEE Journal of Biomedical and Health Informatics*, 2017, pp. 1422–1433.
- [12] W. Chen, D. McDuff, Deepphys: Video-based physiological measurement using convolutional attention networks, in: ECCV, 2018, pp. 349–365.
- [13] X. Chen, J. Cheng, R. Song, Y. Liu, R. Ward, Z.J. Wang, Video-based heart rate measurement: Recent advances and future prospects, in: *IEEE Transactions on Instrumentation and Measurement*, 2018, pp. 3600–3615.
- [14] O.M. Parkhi, A. Vedaldi, A. Zisserman, Deep face recognition, in: BMVC, 2015, pp. 41.1–41.12.
- [15] G.-S. Hsu, A. Ambikapathi, M.-S. Chen, Deep learning with time-frequency representation for pulse estimation from facial videos, in: *IJCB*, 2017, pp. 383–389.
- [16] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, A.C. Berg, Ssd: Single shot multibox detector, in: ECCV, Springer, 2016, pp. 21–37.
- [17] S. Ren, X. Cao, Y. Wei, J. Sun, Face alignment at 3000 fps via regressing local binary features, in: CVPR, 2014, pp. 1685–1692.
- [18] Y. Liu, A. Jourabloo, W. Ren, X. Liu, Dense face alignment, in: ICCV Workshop, 2017, pp. 1619–1628.
- [19] G.-S. Hsu, C.-H. Hsieh, Cross-pose landmark localization using multi-dropout framework, in: *IJCB*, 2017, pp. 390–396.
- [20] A. Asthana, S. Zafeiriou, S. Cheng, M. Pantic, Robust discriminative response map fitting with constrained local models, in: CVPR, 2013.
- [21] F.D. la Torre, W.-S. Chu, X. Xiong, F. Vicente, J. Ding, J. Cohn, Intraface, in: *IEEE Int. Conf. Auto Face and Gesture Recog. (FG)*, 2015.
- [22] S. Wu, M. Kan, Z. He, S. Shan, X. Chen, Funnel-structured cascade for multi-view face detection with alignment-awareness, *Neurocomputing* 221 (2017) 138–145.
- [23] S. Yang, P. Luo, C.C. Loy, X. Tang, Wider face: A face detection benchmark, in: CVPR, 2016, pp. 5525–5533.
- [24] M. Mathias, R. Benenson, M. Pedersoli, L.V. Gool, Face detection without bells and whistles, in: ECCV, 2014, pp. 720–735.
- [25] J. Redmon, A. Farhadi, Yolo9000: Better, faster, stronger, arXiv preprint arXiv:1612.08242, 2016.
- [26] C. Sagonas, G. Tzimiropoulos, S. Zafeiriou, M. Pantic, 300 faces in-the-wild challenge: the first facial landmark localization challenge, in: CVPRW, 2013, pp. 397–403.
- [27] X. Xiong, F. De la Torre, Supervised descent method and its applications to face alignment, in: CVPR, 2013, pp. 532–539.
- [28] M.V. den Bergh, X. Boix, G. Roig, B. de Capitani, L.V. Gool, Seeds: Superpixels extracted via energy-driven sampling, in: ECCV'12 Proceedings of the 12th European conference on Computer Vision, 2012, pp. 13–26.
- [29] J.-W. Kantelhardt, E.K. Bunde, H.A. Rego, S. Havlin, A. Bunde, Detecting long-range correlations with detrended fluctuation analysis, *Physica A* 295 (3) (2001) 441–454.
- [30] R. Hardstone, S.-S. Poil, G. Schiavone, R. Jansen, V.V. Nikulin, H.D. Mansvelder, K. Linkenkaer-Hansen, Detrended fluctuation analysis: a scale-free view on neuronal oscillations, *Frontiers in Physiology* 6 (450) (2012).
- [31] T.C. O'Haver, Single processing: Differentiation, <https://terpconnect.umd.edu/toh/spectrum/Differentiation.html> (May 2008).
- [32] X. Niu, H. Han, S. Shan, X. Chen, Rhythmnet: End-to-end heart rate estimation from face via spatial-temporal representation, *IEEE Transactions on Image Processing* (2019).

- [33] Y. Jia, E. Shelhamer, J. Donahue, S. Karayev, J. Long, R. Girshick, S. Guadarrama, T. Darrell, Caffe: Convolutional architecture for fast feature embedding, arXiv preprint arXiv:1408.5093, 2014.
- [34] M.Z. Poh, D.J. McDuff, R.W. Picard, Advancements in noncontact, multi-parameter physiological measurements using a webcam, *IEEE Transactions on Biomedical Engineering* 58 (1) (2011) 7–11.
- [35] W. Wang, A.C. den Brinker, S. Stuijk, G. de Haan, Algorithmic principles of remote ppg, *IEEE Transactions on Biomedical Engineering* 64 (7) (2016) 1479–1491.
- [36] W. Chen, D. McDuff, Deepphys: Video-based physiological measurement using convolutional attention networks, in: *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 349–365.



GEE-SERN (JISON) HSU received the dual M.S. degree in electrical and mechanical engineering and his Ph.D. in mechanical engineering from the University of Michigan, Ann Arbor, in 1993 and 1995, respectively. From 1995 to 1996, he was a Post-Doctoral Fellow with the University of Michigan. From 1997 to 2000, he was a Senior Research Staff with the National University of Singapore. In 2001, he joined Penpower Tech., where he led research on face recognition and intelligent video surveillance. His team at Penpower Technology was a recipient of the Best Innovation and Best Product Award at the SecuTech Expo for 3 consecutive years (2005–2007). In 2007, he joined the Department of Mechanical Engineering, National Taiwan University of Science and Technology (NTUST), where he is now an associate professor. His research interests are computer vision and pattern recognition. He received best paper awards in ICMT 2011, CVGIP 2013, CVPRW 2014, ARIS 2017 and CVGIP 2018. He is a senior member of IEEE and IAPR.



RUI-CANG Xie received the B.S. degree from the Department of Photonics in Feng Chia University, Taichung, Taiwan, in 2016. He is currently pursuing the M. S. degree in mechanical engineering at the NTUST, Taiwan. His research interest focuses on deep learning and face recognition.



ArulMurugan Ambikapathi received the B.E. degree in electronics and communication engineering from Bharathidasan University, India, in 2003, the M.E. degree in communication systems from Anna University, Chennai, India, in 2005, and the Ph.D. degree from the Institute of Communications Engineering (ICE), National Tsing Hua University (NTHU), Hsinchu, Taiwan, in 2011. He is currently a Scientist at Deep Learning 2.0, Institute of Infocomm Research, a research wing of Agency for Science, Technology, and Research, Singapore. He was a Team lead and Senior Algorithm Engineer at Utechzone Co. Ltd., Taipei, Taiwan, from Sep. 2014 to June 2018. He was a Postdoctoral Research Fellow with ICE, NTHU, from Sep. 2011 to Aug. 2014. His research interests are in computer vision, deep learning theories and applications, and hyperspectral and biomedical image analysis. Dr. Ambikapathi was the recipient of Gold and Silver medals for academic excellence in his B.E and M.E programs, respectively. He was the recipient of the NTHU Outstanding Student Scholarship award for two consecutive years (2009 and 2010). He was a co-recipient of the IEEE Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing 2011, Best Paper Award. He was awarded the Best Ph.D Thesis Award from IEEE GeoScience and Remote Sensing Society, Taipei Chapter.



Kae-Jy Chou received the Ph.D. degree in EECS from the University of Michigan, Ann Arbor, in 1993. He is currently a senior research scientist at the Artificial Vision Laboratory, National Taiwan University of Science and Technology. He was the chief technology officer in Glead Electronics Co. Ltd from 2015–2019. His research interests include computer vision, deep learning, wireless communication and Internet of Things.